

Stationary Distribution

- The distribution we end up with $t=\text{infinity}$ is called the **stationary distribution** P_∞ of the chain.

$$P_\infty(X) = P_{\infty+1}(X) = \sum_x P(X|x)P_\infty(x)$$

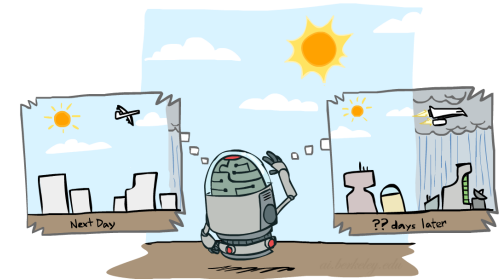
- For most chains, influence of the initial distribution gets less and less over time.
- The distribution we end up in is **independent of the initial distribution**.



Example: Stationary Distribution

- From initial observation of **sun**

$$\begin{array}{ccccccc} \left\langle \begin{array}{c} 1.0 \\ 0.0 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.9 \\ 0.1 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.84 \\ 0.16 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.804 \\ 0.196 \end{array} \right\rangle & \longrightarrow & \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\ P(X_1) & P(X_2) & P(X_3) & P(X_4) & & P(X_\infty) \end{array}$$



- From initial observation of **rain**

$$\begin{array}{ccccccc} \left\langle \begin{array}{c} 0.0 \\ 1.0 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.3 \\ 0.7 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.48 \\ 0.52 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.588 \\ 0.412 \end{array} \right\rangle & \longrightarrow & \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\ P(X_1) & P(X_2) & P(X_3) & P(X_4) & & P(X_\infty) \end{array}$$

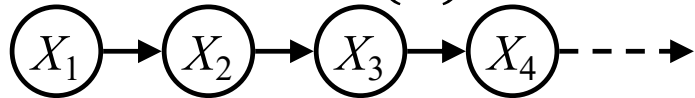
- From yet another initial distribution $P(X_1)$:

$$\begin{array}{ccccccc} \left\langle \begin{array}{c} p \\ 1-p \end{array} \right\rangle & & \dots & & \longrightarrow & \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\ P(X_1) & & & & & P(X_\infty) \end{array}$$

X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

Example: Stationary Distribution

- Question: What's $P(X)$ at time $t = \text{infinity}$?



$$P_{\infty}(\text{sun}) = P(\text{sun}|\text{sun})P_{\infty}(\text{sun}) + P(\text{sun}|\text{rain})P_{\infty}(\text{rain})$$

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$$P_{\infty}(\text{sun}) = 0.9P_{\infty}(\text{sun}) + 0.3P_{\infty}(\text{rain})$$

$$P_{\infty}(\text{rain}) = 0.1P_{\infty}(\text{sun}) + 0.7P_{\infty}(\text{rain})$$

$$P_{\infty}(\text{sun}) = 3P_{\infty}(\text{rain})$$

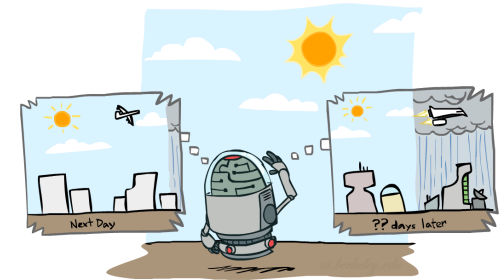
$$P_{\infty}(\text{rain}) = 1/3P_{\infty}(\text{sun})$$

Also: $P_{\infty}(\text{sun}) + P_{\infty}(\text{rain}) = 1$



$$P_{\infty}(\text{sun}) = 3/4$$

$$P_{\infty}(\text{rain}) = 1/4$$



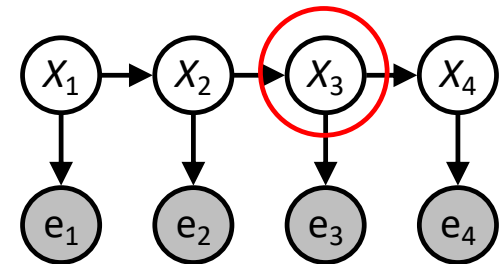
X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
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rain	sun	0.3
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Smoothing

- $P(X_k | e_{1:t})$: the posterior distribution of the state at some past time k given a complete sequence of observations from 1 to t .

$$\begin{aligned} \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:t}) &= \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}, \mathbf{e}_{k+1:t}) \\ &= \alpha \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k, \mathbf{e}_{1:k}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:k}) \\ &= \alpha \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k) \quad (\text{using conditional independence}) \\ &= \alpha \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t}. \end{aligned}$$

- $f_{1:k}$ - **forward** filtering from 1 to k ;
- $b_{k+1:t}$ - **backward** message from later evidences to update X_k .



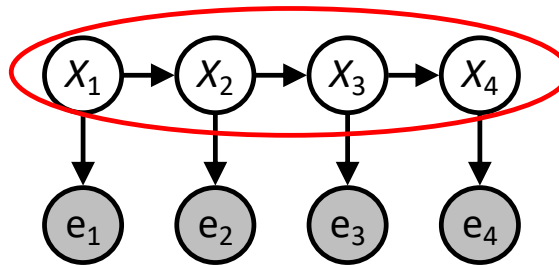
Smoothing

$$\begin{aligned}\mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:t}) &= \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}, \mathbf{e}_{k+1:t}) \\ &= \alpha \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k, \mathbf{e}_{1:k}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:k}) \\ &= \alpha \mathbf{P}(\mathbf{X}_k | \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k) \quad (\text{using conditional independence}) \\ &= \alpha \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t} .\end{aligned}$$


$$\begin{aligned}\mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k) &= \sum_{\mathbf{x}_{k+1}} \mathbf{P}(\mathbf{e}_{k+1:t} | \mathbf{X}_k, \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} | \mathbf{X}_k) \quad (\text{conditioning on } \mathbf{X}_{k+1}) \\ &= \sum_{\mathbf{x}_{k+1}} P(\mathbf{e}_{k+1:t} | \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} | \mathbf{X}_k) \quad (\text{by conditional independence}) \\ &= \sum_{\mathbf{x}_{k+1}} P(\mathbf{e}_{k+1}, \mathbf{e}_{k+2:t} | \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} | \mathbf{X}_k) \\ &= \sum_{\mathbf{x}_{k+1}} \underbrace{P(\mathbf{e}_{k+1} | \mathbf{x}_{k+1})}_{\text{sensor model}} \underbrace{P(\mathbf{e}_{k+2:t} | \mathbf{x}_{k+1})}_{\text{recursion}} \underbrace{\mathbf{P}(\mathbf{x}_{k+1} | \mathbf{X}_k)}_{\text{transition model}} ,\end{aligned}$$

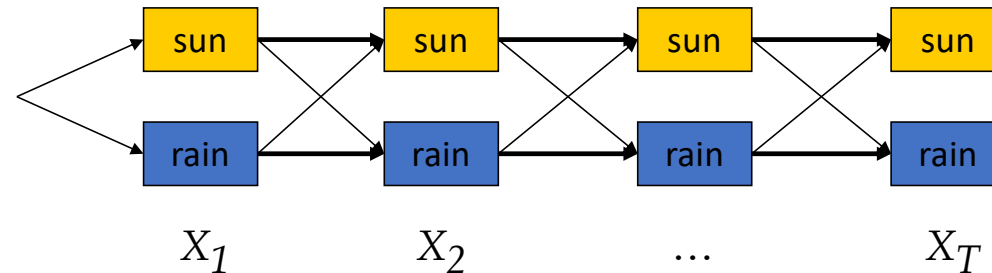
Inference for Most Likely Explanation

- Most likely explanation: $\arg \max_{x_{1:t}} P(x_{1:t} | e_{1:t})$
 - Remember HMM is defined by:
 - Initial distribution: $P(X_1)$
 - Transitions: $P(X_t | X_{t-1})$
 - Emissions: $P(E_t | X_t)$



Most Likely Explanation = Most Probable Path

- Each sequence of states is a **path** through the graph whose nodes are the possible states at each time step.



- The most likely **path to x_{t+1}** consists of the most likely **path to x_t** , plus a **transition from x_t to x_{t+1}** :

$$m_{1:t+1} = \max_{X_{1:t}} P(x_{1:t}, X_{t+1}, e_{1:t+1})$$

Viterbi Algorithm

$$\begin{aligned}
 \mathbf{m}_{1:t+1} &= \max_{\mathbf{x}_{1:t}} \mathbf{P}(\mathbf{x}_{1:t}, \mathbf{X}_{t+1}, \mathbf{e}_{1:t+1}) = \max_{\mathbf{x}_{1:t}} \mathbf{P}(\mathbf{x}_{1:t}, \mathbf{X}_{t+1}, \mathbf{e}_{1:t}, e_{t+1}) \\
 &= \max_{\mathbf{x}_{1:t}} \mathbf{P}(e_{t+1} | \mathbf{x}_{1:t}, \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \mathbf{P}(\mathbf{x}_{1:t}, \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \\
 &= \mathbf{P}(e_{t+1} | \mathbf{X}_{t+1}) \max_{\mathbf{x}_{1:t}} \mathbf{P}(\mathbf{X}_{t+1}, | \mathbf{x}_t) P(\mathbf{x}_{1:t}, \mathbf{e}_{1:t}) \\
 &= \mathbf{P}(e_{t+1} | \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1}, | \mathbf{x}_t) \max_{\mathbf{x}_{1:t-1}} P(\mathbf{x}_{1:t-1}, \mathbf{x}_t, \mathbf{e}_{1:t})
 \end{aligned}$$

Recursion

$\mathbf{m}_{1:t}$

Compared to the forward algorithm for filtering:

$$P(X_t | e_{1:t}) = \alpha P(e_t | X_t) \sum_{x_{t-1}} P(X_t | x_{t-1}) P(x_{t-1} | e_{1:t-1})$$

Exact and Approximate Inferences in HMMs

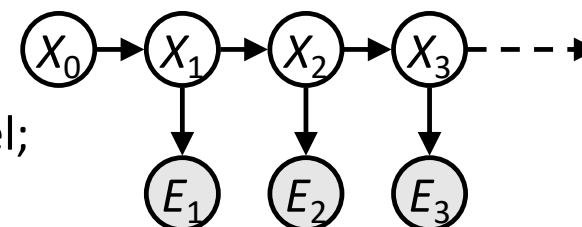
- Exact inference: computes the exact probabilities and state sequences.
 - Forward-Backward algorithm, Viterbi algorithm.
 - $O(N^2T)$, where N is the number of states and T is the length of the sequence.
- Approximate inference: estimates the probabilities and state sequences when exact computation is infeasible.
 - E.g., T is very long.

Likelihood Weighting for HMMs

- Likelihood weighting is a type of importance sampling:
 - Samples are drawn from a proposal distribution;
 - Each sample is weighted by its likelihood under the target distribution;
 - Commonly used in static Bayesian networks.
- For HMMs:
 - Generate state sequences and corresponding observations;
 - Adjust weights to reflect how well the sample explains the observed sequence;
 - Helps estimate hidden states when exact inference is too costlythe process involves generating state sequences and observations while adjusting weights to reflect the probability of the observed sequence given the state sequence.

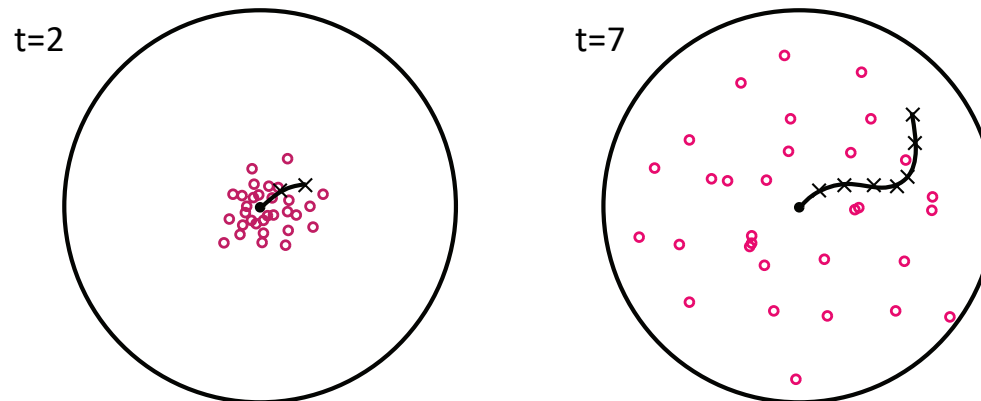
Likelihood Weighting for HMMs

- Sample the current state X_t :
 - Based on the previous state X_{t-1} and the transition model;
- Evaluate the observation likelihood:
 - Compute the likelihood using the observation model;
- Update the sample weight:
 - Multiply the current weight by the likelihood (initial weight = 1);
- Repeat for each time step:
 - Continue sampling and weighting to build a full sequence.



We Need a New Idea!

- Reason of the failure: Sample state trajectories go into very low-probability regions.
- Solution: kill the bad ones, make more of the good ones.
 - **Resampling**: The population of samples stays in high-probability region.

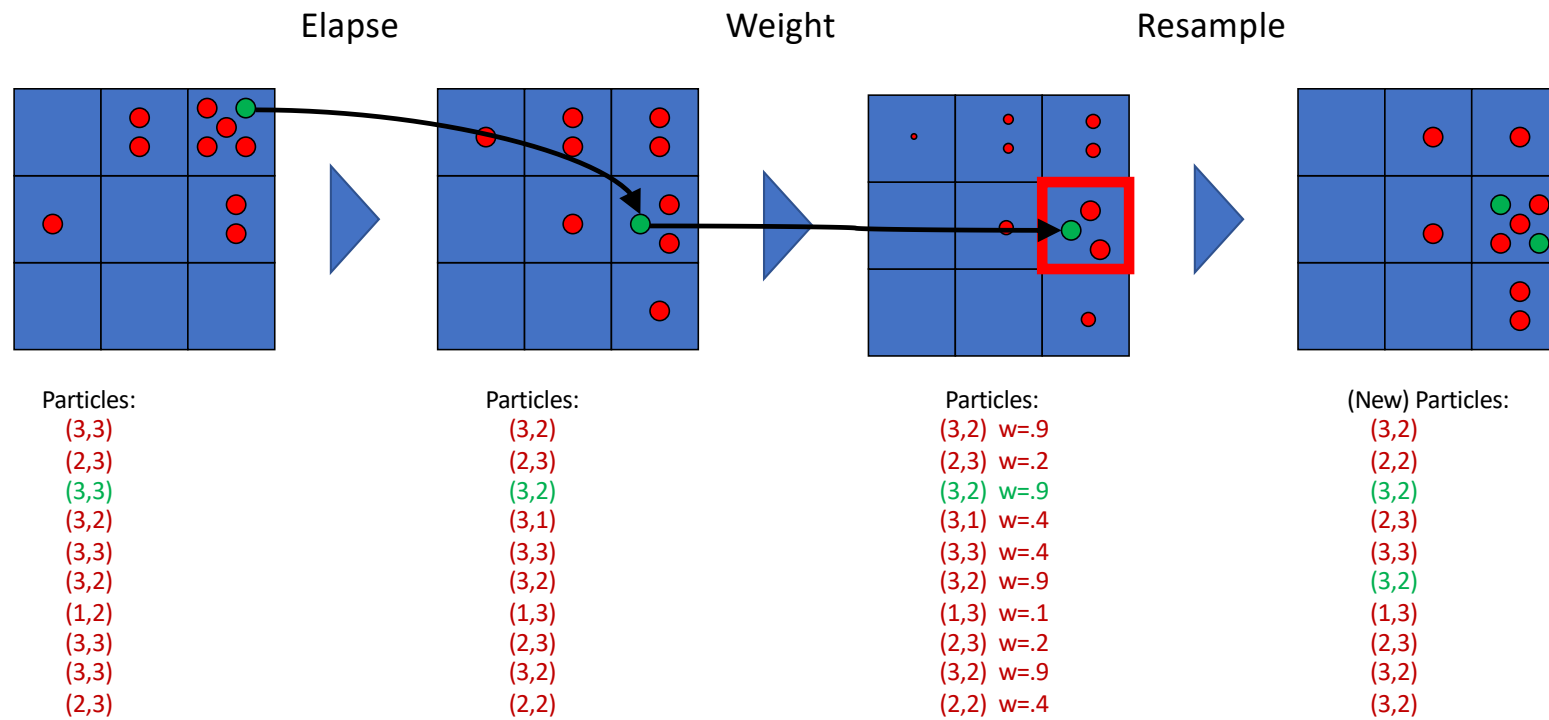


Particle Filtering

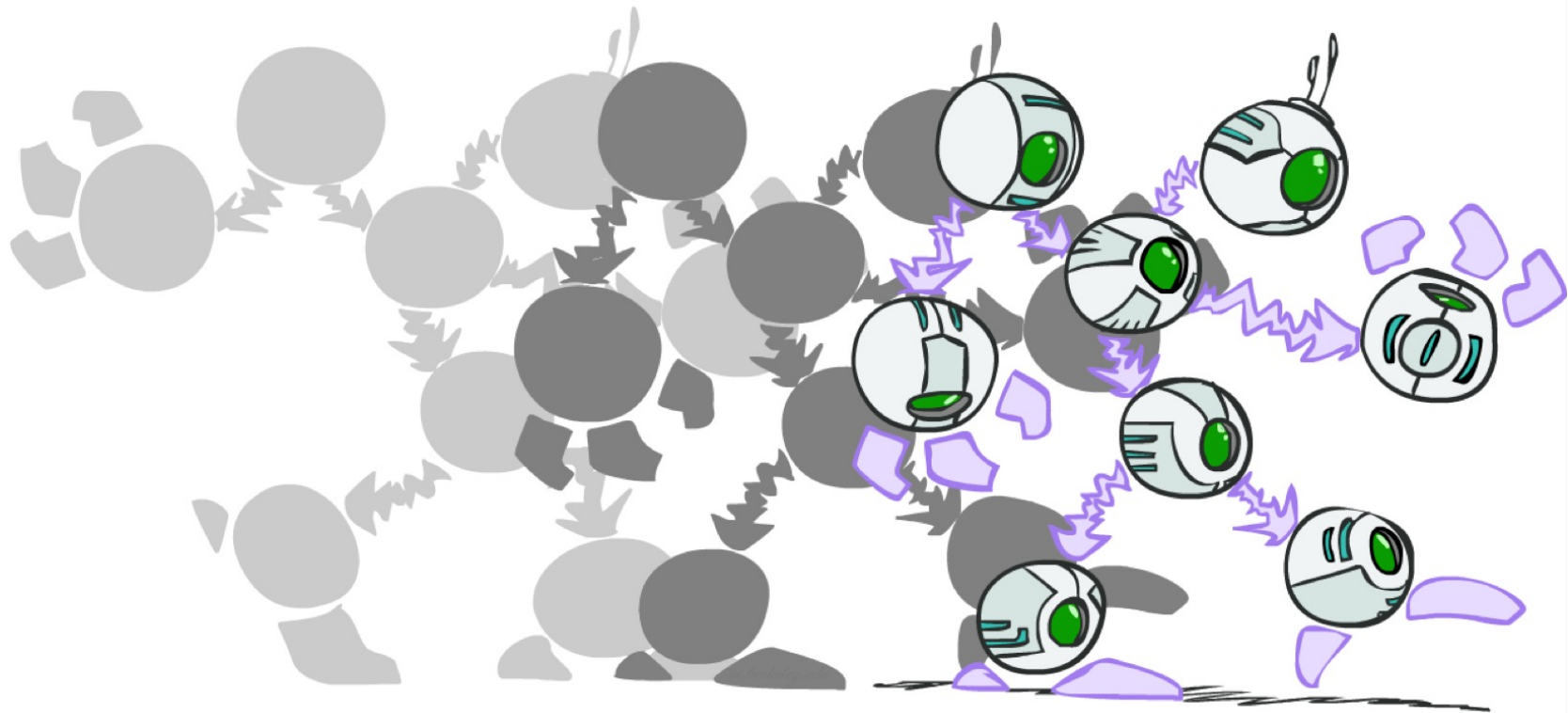
- Particle filtering: A sequential Monte Carlo method for approximating belief states in dynamic models like HMMs.
 - Handles nonlinear and non-Gaussian models;
 - Scales better than exact inference for long sequences;
 - Maintains a focused sample set through resampling.
- How the learn the concept of “Particle Filtering”?

Recap: Particle Filtering

- Particles: track samples of states rather than an explicit distribution.

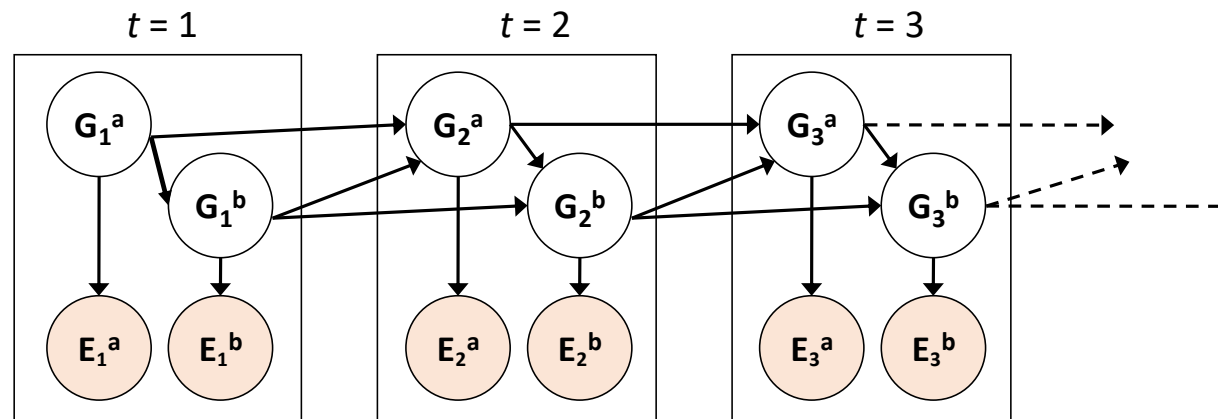


Dynamic Bayesian Network (DBN)



Dynamic Bayesian Network (DBN)

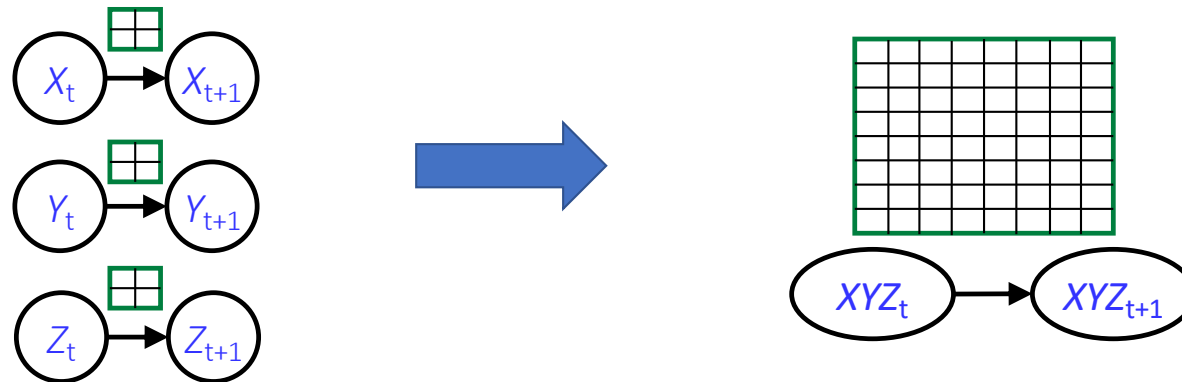
- DBN:
 - Designed to track multiple variables over time;
 - Integrates multiple sources of evidence across time steps;
 - Core idea: Repeat a fixed Bayesian network structure at each time step.



Variables from time t can condition on those from $t-1$.

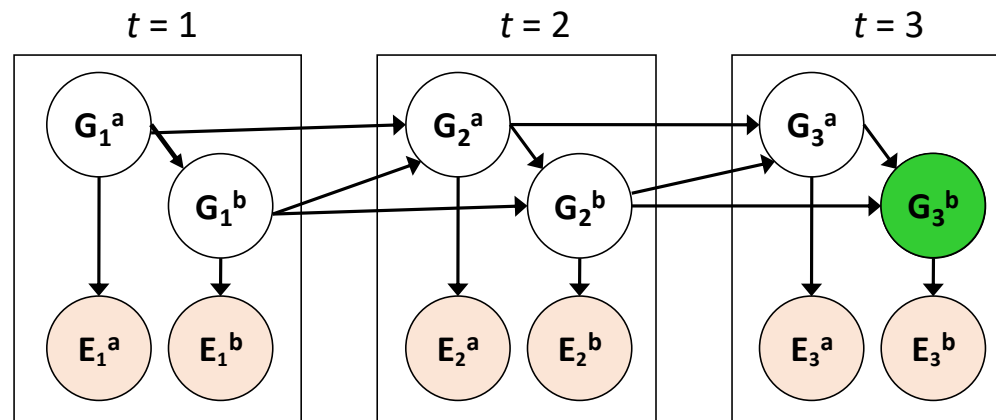
DBN and HMM

- HMM's hidden state is a composite variable formed by combining all DBN state variables.
 - Every HMM can be viewed as a single-variable DBN.
 - DBNs exploit sparse dependencies to reduce complexity.



Exact Inference in DBNs

- Goal: Compute $P(X_T | e_{1:T})$, the belief at the final time step.
 - Unroll the DBN for T time steps;
 - Apply variable elimination across time;



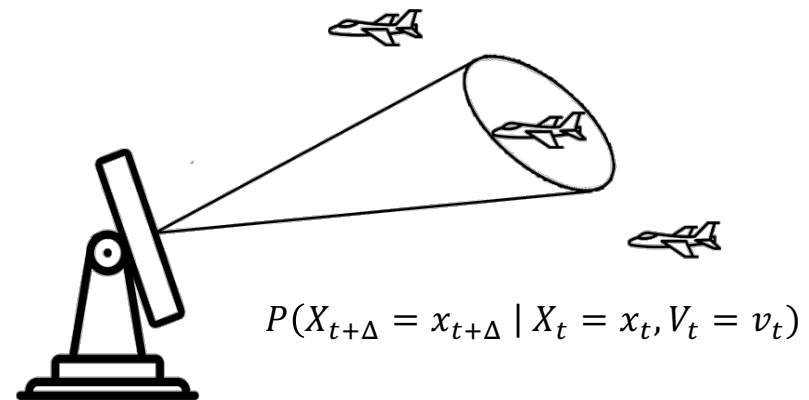
- Online belief update: Retain only current factors for the next update.
 - This enables real-time inference with minimal memory usage.

Approximate Inference: DBN Particle Filters

- **Initialize:** Generate prior samples for the $t=1$ Bayes net;
- **Elapse time:** Sample a successor for each particle;
- **Observe:** Weight each **entire** sample by the likelihood of the evidence conditioned on the sample;
- **Resample:** Select prior samples (tuples of values) in proportion to their likelihood.

Kalman Filter

- Motivation:
 - HMMs assume discrete hidden states and observations;
 - But many real-world systems involve continuous variables.
- Kalman Filter:
 - A continuous-state version of HMM;
 - Assumes linear dynamics and Gaussian noise;
 - Enables efficient exact inference using matrix operations.



Example:

- The location of the plane is a continuous state variable.
- The reading of the radar is a continuous evidence variable.

Kalman Filter

- We continue with discretized time steps t ;
- Can infer the state variable from t to $t + 1$;

$$\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}) = \int_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t) P(\mathbf{x}_t | \mathbf{e}_{1:t}) d\mathbf{x}_t$$

- Can further update the state belief given evidence at $t + 1$.

$$\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) = \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}) \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t})$$

Additional Resource

- Illustration of HMMs in Chinese
 - <https://zhuanlan.zhihu.com/p/497045631>
- Kalman filter
 - https://en.wikipedia.org/wiki/Kalman_filter
 - <https://www.kalmanfilter.net/default.aspx>

Thank You!